

Lab: Collective Problem-Solving

Group Activity — Cognition

Objective

Experience and analyze collective cognition in action by working through real crochet problems as a group and observing how distributed knowledge combines to produce solutions.

Materials

- Each member's current crochet project
- 2-3 "mystery swatches" — small pieces of crochet that contain deliberate problems (gauge inconsistency, a missed increase, an unfamiliar stitch pattern)
- Index cards for writing hypotheses
- Pens
- A timer

Duration

50-60 minutes

Preparation

Before the meeting, one or two members prepare "mystery swatches" — small crochet samples that have something to figure out. Ideas: - A swatch with a deliberate error (dropped stitch, miscounted repeat) - A swatch in an unfamiliar stitch that the group must identify - A swatch where the yarn choice creates an unexpected result (unusual drape, curling, splitting)

Activity Steps

Step 1: The "Has Anyone Tried This?" Round (10 minutes)

Go around the circle. Each person poses a real question — something they are genuinely uncertain about in their current project or crochet knowledge. It does not have to be a problem; it can be curiosity. Examples: - "Has anyone blocked acrylic successfully?" - "What is the best way to join granny squares?" - "How do you keep count when you are chatting?"

After each question, let the group respond. Notice: Does someone have an answer? Do multiple people contribute partial answers? Does anyone say "I do not know either, but I wonder if...?"

After the round, discuss: How much of the circle's knowledge surfaced through these simple questions? Was anything surprising?

Step 2: Mystery Swatch Diagnosis (15 minutes)

Present the mystery swatches one at a time. For each swatch, the group works through the problem-solving process:

1. **Examine:** Pass the swatch around. Everyone looks, touches, holds it up to the light.
2. **Hypothesize:** Each person writes a hypothesis on an index card about what is going on with the swatch. (Do not discuss yet.)
3. **Share:** Read hypotheses aloud. Notice where they agree and differ.
4. **Evaluate:** As a group, compare hypotheses against the evidence of the swatch. Which hypotheses fit the evidence? Which do not?
5. **Converge:** Try to reach a consensus explanation. If you cannot, that is interesting too — record the competing hypotheses.

After each swatch, reflect: Whose knowledge was crucial? Did the answer require combining knowledge from multiple people?

Step 3: Nonverbal Problem-Solving (10 minutes)

Choose a real crochet challenge from Step 1 that lends itself to demonstration. The rule: **solve it without speaking**. One person demonstrates the technique or solution using only their hands, their hook, and their yarn. The person with the question watches and tries to follow.

After the demonstration, discuss: - Was the nonverbal explanation clear? - What information came through easily? What was hard to convey without words? - Did the demonstrator adjust their showing based on the learner's visible confusion (nonverbal feedback)?

Step 4: Map the Group's Knowledge (10 minutes)

On a shared sheet of paper, create a quick "knowledge map." List major crochet domains across the top (e.g., Basic Stitches, Color Work, Garment Construction, Amigurumi, Lace, Finishing Techniques). Each member places their initials under the domains where they have knowledge.

Look at the map: - Where is the group's knowledge concentrated? - Where are the gaps — domains where no one has expertise? - How does this map help you see the circle's collective cognitive capacity?

Reflection Prompts

- During the mystery swatch exercise, was the group's diagnosis better than what any individual could have produced alone?
- How did it feel to combine your partial knowledge with others'?
- Were there moments of genuine surprise — when someone's contribution changed the group's understanding?
- How does the circle's cognitive capacity compare to what you would have access to working alone?

Extension

Start a "Circle Knowledge Log" — a shared notebook or document where the group records solutions to problems, tips discovered, and techniques learned. This externalizes the circle's distributed cognition, making it accessible even when not all members are present. Revisit the log periodically and discuss how the group's shared knowledge has grown.